

Joshua Dinn

QUANTITATIVE RESEARCH & TRADING

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EDUCATION

Monash University, Melbourne, Australia

Mar 2023 – Oct 2027

Bachelor of Engineering (Software Engineering) and Bachelor of Commerce (Econometrics)

WAM: 79.97 (Distinction) | GPA: 3.48 / 4.0 | 100% in FIT2107 (Software Quality & Testing)

PROJECTS

Algorithmic Trading System for S&P 500 ETF

May 2025

- Built and backtested a trading system using Hidden Markov Models to classify market regimes for SPY.
- Used RSI, ATR, and volatility signals to drive position sizing and trade execution.
- Deployed it on an AWS Linux server with the Alpaca API for automated paper trading.

Volatility Forecasting with a GARCH Model

Apr 2025

- Built a volatility forecasting pipeline for the S&P 500 using a GARCH(1,1) model fitted by maximum likelihood on a decade of daily log returns.
- Validated the fit with Ljung-Box and ARCH-LM residual diagnostics, then forecast conditional variance 100 days ahead, automating data processing, evaluation, and reporting in Python.

OLS vs MLE: Monte Carlo Study of Estimator Performance

Ongoing

- Designing a Monte Carlo study (Python) comparing OLS against correctly specified maximum likelihood estimators when the classical linear model assumptions fail: heteroscedasticity, skewed and fat-tailed errors, and binary or count outcomes.
- Measuring bias, relative MSE, and empirical coverage of nominal 95% confidence intervals across sample sizes from 25 to 2,000, with thousands of replications per design, to show where standard inference stops being trustworthy.
- Deriving and implementing the Poisson regression MLE from scratch, solving the score equations with Newton-Raphson (IRLS) instead of calling a library fitter.

JackTen: No-Limit Hold'em Poker Bot

Ongoing

- Building a poker decision engine from first principles in Python: hand evaluation and equity first, then the expected value, pot odds, and minimum defence frequency calculations that drive every action.
- Modelling opponents with Bayesian range updates from observed actions, combined with tracked statistics (VPIP, PFR, 3-bet%) to move from a balanced baseline strategy to exploitative play.

EXPERIENCE

Collingwood Football Club, Data & Analytics Intern

Mar 2026 – Jun 2026

- Built a membership churn prediction model in XGBoost on historical transaction and engagement data, engineering the feature pipelines and validating the model before it went in front of the commercial team.
- Used SHAP analysis to explain which behaviours drove the predictions, and built an interactive dashboard breaking churn down by segment.

Monash DeepNeuron, Optimised Computing Member

Sep 2025 – Present

- Software team member on HarvestAI, an autonomous fruit-picking robotic arm, training an object detection model for real-time strawberry detection and ripeness classification on HPC-accelerated runs.
- Building an on-device vector database with HNSW indexing and approximate nearest-neighbour search for a low-latency retrieval pipeline.
- Teaching Assistant for an 8-week C++ and accelerated computing workshop on GPU programming, CUDA, and HPC, run with NVIDIA, the Pawsey Supercomputing Centre, and NCI. NVIDIA CUDA certified.

Outlier AI, AI Trainer (Mathematics)

Jun 2024 – Dec 2025

- Reviewed and corrected AI-generated mathematical solutions written in LaTeX to improve accuracy and reasoning quality.
- Wrote detailed feedback on computational and logical errors across advanced problem sets.

TMJ Tutoring, Mathematics Tutor

Feb 2023 – Present

- Tutor high school students in mathematics, focusing on exam preparation and breaking hard problems into steps they can follow.

SKILLS

Languages: Python, C++, R, MATLAB, SQL

Econometrics & Statistics: Maximum likelihood estimation, Monte Carlo simulation, GARCH and volatility modelling, Hidden Markov Models, Bayesian updating, panel data and fixed effects, asymptotic inference

Quantitative Tools: pandas, NumPy, SciPy, statsmodels, OxMetrics, backtesting, Alpaca API, AWS, Linux, Git

Interests: Financial econometrics, statistical modelling, game theory, machine learning